

Lepton Mass Ratios from UBT Constants

Structural coincidences: $h_\tau \approx \sqrt{2}\alpha$, $m_\tau/m_e \approx 2\pi/\Omega_\eta$, $m_\tau/m_\mu \approx 4\alpha/\Omega_\eta$

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Abstract

We document three structural coincidences between charged lepton masses and UBT-derived constants. All three involve only $\alpha^{-1} = 137.036$ `[[L1], alpha.l1.proof.tex]` and $\Omega_\eta = \frac{1}{24} - \frac{1}{8\pi}$ `[[L0]]`:

$$\begin{aligned} h_\tau &\approx \sqrt{2}\alpha & (-1.0\%), \\ m_\tau/m_e &\approx 2\pi/\Omega_\eta & (-3.8\%), \\ m_\tau/m_\mu &\approx 4\alpha/\Omega_\eta & (-7.5\%). \end{aligned}$$

Status: `[[MC+]]` — structural evidence. No derivation from $S[\Theta]$ is available; the EW hierarchy problem ($M_{\text{EW}} \ll M_{\text{UBT}}$) must first be resolved. We record these as falsifiable predictions.

1 UBT Constants

$$\alpha_{\text{UBT}}^{-1} = 137.035999177549\dots \quad \text{[[L1]]} \quad (1)$$

$$\Omega_\eta = \frac{1}{24} - \frac{1}{8\pi} = 0.001877928\dots \quad \text{[[L0]]} \quad (2)$$

$$C_Q = 4 \quad \text{[[L1]]} \quad (3)$$

2 Three Structural Coincidences

Observation 2.1 (Tau Yukawa $\approx \sqrt{2}\alpha$ — `[[MC+]]`). The tau-lepton Yukawa coupling $h_\tau = m_\tau\sqrt{2}/v$ satisfies:

$$h_\tau = \frac{m_\tau\sqrt{2}}{v} = \frac{1.7769 \times \sqrt{2}}{246.22} = 0.010215\dots \approx \sqrt{2}\alpha = 0.010320\dots$$

Discrepancy: -1.0% . No free parameter used.

Observation 2.2 (Tau-to-electron ratio — `[[MC+]]`).

$$\frac{m_\tau}{m_e} = \frac{1776.9 \text{ MeV}}{0.511 \text{ MeV}} = 3477.3 \approx \frac{2\pi}{\Omega_\eta} = \frac{2\pi}{0.001878} = 3345.8$$

Discrepancy: -3.8% .

Observation 2.3 (Tau-to-muon ratio — [MC+]).

$$\frac{m_\tau}{m_\mu} = \frac{1776.9 \text{ MeV}}{105.7 \text{ MeV}} = 16.81 \approx \frac{4\alpha}{\Omega_\eta} = \frac{4/137.036}{0.001878} = 15.54$$

Discrepancy: -7.5% .

Lepton mass formula (structural, [MC+]). If $m_\tau \approx v\alpha$ and the ratios hold:

$$m_\tau : m_\mu : m_e \approx v\alpha : \frac{v\Omega_\eta}{C_Q} : \frac{v\Omega_\eta\alpha}{2\pi} = v\alpha : \frac{v\Omega_\eta}{4} : \frac{v\Omega_\eta\alpha}{2\pi}.$$

All ratios involve only α [[L1]] and Ω_η [[L0]]:

$$\begin{aligned} \frac{m_\tau}{m_\mu} &\approx \frac{C_Q\alpha}{\Omega_\eta} = \frac{4\alpha}{\Omega_\eta} = 15.54 \quad (\text{exp. } 16.81, -7.5\%), \\ \frac{m_\tau}{m_e} &\approx \frac{2\pi}{\Omega_\eta} = 3346 \quad (\text{exp. } 3477, -3.8\%). \end{aligned}$$

3 Discussion

Remark 3.1 (Physical interpretation). The formula $m_\tau \approx v\alpha$ suggests the tau Yukawa coupling $h_\tau \approx \sqrt{2}\alpha$ is proportional to the EM coupling. This would arise naturally if the tau mass is generated at one loop in EM, analogously to the anomalous magnetic moment ($g - 2 \sim \alpha/\pi$).

The formula $m_e \approx v\Omega_\eta\alpha/(2\pi)$ suggests the electron mass involves two suppressions: one from Ω_η (eta-mode occupation, a ψ -sector loop factor) and one from $\alpha/(2\pi)$ (EM loop factor).

What is missing.

These coincidences are [[MC+]], not [[L1]]. Three things are needed:

1. **VEV hierarchy** $v = 246 \text{ GeV} \ll M_{\text{UBT}}$: the EW hierarchy problem [[Open]] must first be resolved.
2. **Derivation of $h_\tau = \sqrt{2}\alpha$** from $S[\Theta]$: the tau Yukawa coupling must be shown to equal $\sqrt{2}\alpha$ in the UBT Yukawa sector.
3. **Derivation of generation structure**: why do the three generations correspond to this specific pattern $(\alpha, \Omega_\eta/C_Q, \Omega_\eta\alpha/2\pi)$?

Without resolving the hierarchy problem, these remain structural coincidences without a derivation.

Remark 3.2 (Falsifiability). The predictions $m_\tau/m_e = 2\pi/\Omega_\eta = 3346$ and $m_\tau/m_\mu = 4\alpha/\Omega_\eta = 15.5$ are falsifiable. They are wrong at the 4–8% level, which may indicate: (a) higher-order corrections of order Ω_η or α/π ; (b) the identification is approximate and the true formula differs; (c) coincidence. Distinguishing these requires a derivation from $S[\Theta]$.

References

- [1] D. Jaros, `alpha_l1_proof.tex`, 2026.
- [2] D. Jaros, `layer2_kernel_derivation.tex`, 2026.
- [3] D. Jaros, `higgs_yukawa_status.tex`, 2026.